level configurations via CELLECT. Horizontal and vertical line profiles are depicted in purple and orange, respectively. **b**, Extended view of ground truth-applied maps. From the ground truth of the center location of each cell, we generated a 3-voxel radius binary mask for the 2 frames and multiplied it with intensity maps and confidence maps, respectively. The location of the extended view is indicated with a green box in each original image in (**a**). Overlapped areas with ground truth are shown in red and cyan for intensity and confidence maps, respectively. **c-d**, The impact of level configuration on the model's performance was evaluated, demonstrating that the 4-level configuration used in this study achieved the best performance. This configuration enhances center point distinction through multi-level refinement. Error connections are highlighted in pink, while correct connections are depicted in grey. All evaluation data in this figure were drawn from mskcc-confocal dataset #2. All models were trained using mskcc-confocal dataset #3. Scale bars: 10 μm.